VOLUME 6 SURVEILLANCE

CHAPTER 9 TITLE 14 CFR PART 145 INSPECTIONS

Section 8 Inspect a 14 CFR Part 145 Repair Station's Housing and Facilities

Source Basis:

- **Title 14 CFR Part 43, Maintenance, Preventive Maintenance, Rebuilding, and Alteration.**
- **Title 14 CFR Part 65, Certification: Airmen Other Than Flight Crewmembers.**
- **Title 14 CFR Part 91, General Operating and Flight Rules.**
- **Title 14 CFR Part 121, Operating Requirements: Domestic, Flag, and Supplemental Operations.**
- **Title 14 CFR Part 125, Certification and Operations: Aircraft Having a Seating Capacity of 20 or More Passengers or a Maximum Payload Capacity of 6,000 Pounds or More; and Rules Governing Persons On Board Such Aircraft.**
- **Title 14 CFR Part 135, Operating Requirements: Commuter and On Demand Operations and Rules Governing Persons On Board Such Aircraft.**
- **Title 14 CFR Part 145, Repair Stations.**

6-1773 REPORTING SYSTEM(S). For Title 14 of the Code of Federal Regulations (14 CFR) part 145, use Safety Assurance System (SAS) automation and the associated Data Collection Tools (DCT).

6-1774 OBJECTIVE. This section provides guidance for repair stations during inspection, surveillance, change in location, or adding facilities, per 14 CFR § 145.103.

6-1775 GENERAL.

A. Certificated Repair Station (CRS) Accommodations. The CRS must provide facilities to accommodate equipment, materials, and personnel needed to properly perform maintenance, preventive maintenance, alterations of articles, or the specialized services for the rating it receives. When inspecting the repair station, the aviation safety inspector (ASI) should determine which items apply based on the complexity of the facility and the level of ratings.

NOTE: A repair station applying for a change to its certificate (name, location, or ratings) must submit a request on Federal Aviation Administration (FAA) Form 8310-3, Application for Repair Station Certificate and/or Rating. The change to the certificate may affect the housing requirements or require additional procedures in the Repair Station Manual (RSM)/Quality Control Manual (QCM).

B. Line Maintenance Authorization Locations. A repair station may hold a line maintenance authorization for locations outside of the repair station's fixed location if the repair station already provides housing, as required by 14 CFR § 145.103(a). The repair station may perform maintenance, preventive maintenance, or alterations on articles outside of its housing if it provides suitable facilities that are acceptable to the FAA and meet the requirements of

14 CFR § 145.103(a) so that the work can be done in accordance with the requirements of 14 CFR part 43.

6-1776 REFERENCES, FORMS, AND JOB AIDS.

A. References (current editions):

1) FAA Order 8900.1:

- Volume 1, Chapter 3, Section 1, Responsibilities of Aviation Safety Inspectors.
- Volume 6, Chapter 9, Section 9, Inspect a Part 145 Repair Station's Tools and Equipment.
- Volume 6, Chapter 11, Section 17, Safety Assurance System: Inspect Avionics Test Equipment.
- Volume 6, Chapter 11, Section 20, Evaluate Special Equipment or Test Apparatus.
- Volume 6, Chapter 11, Section 27, Inspect a Nondestructive Inspection Program/Facility.
- Volume 6, Chapter 11, Section 28, Inspect a Repair Facility Accomplishing Composite Maintenance and Repairs.
- Volume 10, Safety Assurance System Policy and Procedures.
- Volume 14, Chapter 1, Section 2, Flight Standards Service Compliance Action Decision Procedure.

2) Advisory Circulars (AC):

- AC 43-15, Recommended Guidelines for Instrument Shops.
- AC 43-214, Repairs and Alterations to Composite and Bonded Aircraft Structure.
- AC 145-9, Guide for Developing and Evaluating Repair Station and Quality Control Manuals.

B. Forms. None.

C. Job Aids. The following job aids can be found in the Dynamic Regulatory System (DRS):

- Nondestructive Inspection for Aviation Safety Inspectors.
- On-Line Job Aid for Evaluating a Repair Facility Conducting Composite Repairs.

NOTE: To find job aids in DRS, in the left menu click "Other Documents," "Reporting Systems and Job Aids," and then "Job Aids."

6-1777 PROCEDURES. Review applicable information, including the Repair Station Certificate manuals/revision, capabilities listing, and operations specifications (OpSpecs) for

accuracy to determine that ratings are appropriate for the work being performed. Before the inspection, the principal inspector (PI) should carefully review the following:

- Title 14 CFR part 145.
- RSM/QCM.
- OpSpecs, to include OpSpecs A101, D100, and D107, if applicable.
- Applicable reports generated in SAS or the Safety Performance Analysis System (SPAS).
- SAS Configuration [Module 1] Vitals Information.
- Responsible Flight Standards office file.

6-1778 CRS HOUSING.

A. Housing General. The ASIs should inspect the repair station facilities for the proper performance of maintenance, preventive maintenance, alterations of articles, or the specialized services consistent with its ratings and limitations. The facilities must include the following:

1) Sufficient workspace and areas for the proper segregation and protection of articles during all maintenance, preventive maintenance, or alterations.

2) Segregated work areas enabling environmentally hazardous or sensitive operations (such as painting, cleaning, nondestructive testing, composite repairs, surface treatments, welding, avionics work, electronic work, and machining) to be done properly and in a manner that does not adversely affect other maintenance or alterations.

3) Suitable racks, hoists, trays, stands, and other segregation means for the storage and protection of all articles undergoing maintenance, preventive maintenance, or alterations.

4) Space sufficient to segregate articles and materials stocked for installation from those articles undergoing maintenance, preventive maintenance, or alterations.

5) Ventilation, lighting, and control of temperature, humidity, and other climatic conditions sufficient to ensure personnel perform maintenance, preventive maintenance, or alterations to the standards this part requires.

NOTE: In most circumstances, the article's maintenance manuals, Component Maintenance Manual (CMM), or Standard Practices Manuals (SPM) provide the recommended environmental conditions.

B. Housing Requirements. ASIs should inspect and review the housing needs of the repair station.

1) A CRS must provide housing for the facilities, equipment, materials, and personnel consistent with its ratings and limitations. Title 14 CFR § 145.103(b) provides a general exception to perform work outside of its housing if it provides suitable facilities that are acceptable to the FAA and meet the requirements of 14 CFR § 145.103(a), so that the work can be done in accordance with the requirements of 14 CFR part 43.

2) Each CRS must have a fixed location where materials, equipment, tools, and data are controlled and stored. While consideration can be given for certain operating situations, if the repair station's scope of work is limited to work that does not require the size and type of housing that the rating requires due to its limitations, the repair station would only need to provide housing sufficient to accommodate its limited scope of work.

3) Repair stations that frequently work away from their fixed location must ensure the housing and facilities are adequate and meet the requirements of the regulations for the ratings and limitations they hold. Their manuals should describe their housing, facilities, equipment, and materials, as required by 14 CFR part 145 subpart C. If a repair station occasionally needs access to house an entire aircraft, the procedures should describe this type of operation. If a hangar is leased temporarily, the procedures should describe how to evaluate a facility prior to performing maintenance. The procedures should describe the use of temporary/portable inflatable tents or shelters, which are approved by aircraft manufacturers as an alternate means to perform those maintenance tasks that need to be under dry/temperate conditions.

4) To be issued a line maintenance authorization, the repair station must show the capability to perform the limited scope of work defined in 14 CFR § 145.3(d) at each location. After demonstrating the capability, the repair station must perform the maintenance in accordance with 14 CFR § 145.205. It is permissible to issue a line maintenance authorization to a repair station that does not have a hangar to house the aircraft listed on the airframe rating. The work performed must meet the requirements of 14 CFR § 145.103(a), so that the work can be done in accordance with the requirements of 14 CFR part 43 and § 145.205.

C. Air Carrier Maintenance. The ASI should review the RSM/QCM to verify procedures for a CRS performing maintenance, preventive maintenance, or alterations for the following certificated operators or carriers:

1) A 14 CFR part 121 or 135 air carrier or commercial operator that has a Continuous Airworthiness Maintenance Program (CAMP). The repair station must follow their program and determine if the housing is sufficient for the work performed.

2) A 14 CFR part 125 operator and the repair station must follow the operator's FAA-approved inspection program. The repair station must determine if the housing is sufficient for the work performed.

3) A foreign air carrier or foreign person operating a U.S.-registered aircraft and the repair station must follow the operator's FAA-approved maintenance/inspection program. The repair station must determine if the housing is sufficient for the work performed.

D. Verify Segregation and Protection of Parts. Verify that each workspace has areas for the proper segregation and protection of parts and subassemblies during all phases of maintenance, preventive maintenance, or alterations. Verify the following:

1) The differences between serviceable and unserviceable components, parts, and materials must be clearly distinguishable throughout each process.

2) Repair stations may accomplish this with suitable racks, hoists, trays, stands, or other means of segregation for storing and protecting all articles undergoing maintenance, preventive maintenance, or alterations.

3) Repair stations should segregate environmentally hazardous or sensitive operations so that they do not adversely affect other maintenance or alteration of articles or activities. Examples of these operations are avionics work, electrostatic discharge (ESD), battery maintenance, painting, surface treatments, composite repairs, nondestructive testing, cleaning, welding, and machining.

4) If the facility deals in non-aircraft parts, materials, or maintenance activities outside those normally performed at the repair station, the facility should segregate the aircraft function from other functions to preclude a repair station using unapproved parts or materials on an aircraft.

5) Repair stations must segregate articles and materials stocked for installation from articles and materials undergoing maintenance, preventive maintenance, or alteration.

E. Determine Adequacy of Environmental Conditions. The repair station must have sufficient ventilation, lighting, and control of temperature, humidity, and other climate conditions to ensure personnel are not adversely affected by the environmental conditions when performing maintenance, preventive maintenance, or alterations. In addition to reasonable heating, air conditioning, and lighting requirements, verify the following maintenance environmental conditions:

1) Instrument shops and composite layup and clean rooms are environmentally and operationally controlled per the Original Equipment Manufacturer (OEM) or an FAA-approved repair process. This may include measurement of air cleanliness, such as clean room requirements.

2) Storage areas include proper storage for such items as flammables, sealants, chemicals, carcinogens, tires, or tooling, and are environmentally controlled in regard to temperature and humidity.

3) Lighting is adequate for the type of processes performed in each area.

4) While physically inspecting the repair station, verify that facility diagrams and descriptions in the RSM are accurate. This includes any facilities used for spray painting, avionics, engine or airframe repair, or any other work that would have special requirements. Pay close attention to specific information detailed in the manual, such as:

- The type of heating,
- Lighting,
- Equipment location,
- Electrical,
- Compressed air outlets, and
- Ventilation.

F. Corporate Safety Culture. Repair stations must create a safe working environment that will help prevent personnel injury and damage to customer property. The housing and facilities should provide adequate security and fire protection. The PI should review the repair station's safety procedures. The PI should also consider that poor housekeeping or improper maintenance of safety devices (such as eyewash stations, material safety data sheets (MSDS), and fire extinguishers) is a good indicator of the repair station's corporate culture.

NOTE: This inspection focuses on how the repair station follows its safety policies and procedures. Outside of the PI's jurisdiction are safety, fire, and health rules, codes, and regulations, which vary from one state or county to another.

G. Inspect General Housekeeping. Inspect the repair station to determine that general housekeeping will not contaminate component parts and subassemblies undergoing maintenance. Repair stations should maintain all shops and hangars in a clean and orderly manner.

1) Methods should include procedures to prevent components or materials from damage from ESD. Materials should be packaged, handled, and protected in accordance with requirements for handling ESDs.

2) Testing facilities and test rigs should meet the component manufacturers' service criteria and contaminant prevention requirements. Scheduled maintenance of the testing rigs should document the cleanliness levels, filter replacements, calibration, etc.

6-1779 TASK OUTCOMES.

A. Conduct Debriefing. Brief the certificate holder (CH) on the results. Discuss all deficiencies, CH corrective actions, and FAA actions. The ASI can find instructions for conducting briefings in Volume 1, Chapter 3, Section 1.

B. Compliance and Enforcement Action. If safety issues and/or regulatory noncompliance are identified, follow the process contained in Volume 14, Chapter 1, Section 2 to determine the appropriate FAA compliance or enforcement action.

C. Complete the Task. Follow Volume 10 guidance for completion of SAS Activity Recording (AR) or the DCT.

6-1780 FUTURE ACTIVITIES. Follow Volume 10 to plan future risk-based surveillance in SAS.

RESERVED. Paragraphs 6-1781 through 6-1794.